

The Impact of GDP on Renewable Energy Production

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This report explores the correlation between GDP and renewable energy production. It is found that although there is a correlation between GDP and renewable energy production, the greatest determiner is political beliefs and public opinion.

I. INTRODUCTION

This report aims to examine how the Gross Domestic Product (GDP) of a country impacts the production of renewable energy within that country. To do this, the report looks at regions of the world rather than specific countries.

This report requires an analytical approach to enable a conclusion to be reached. As greenhouse emissions continue to increase, there is a greater incentive than ever to increase renewable energy production as the effects of climate change worsen.[1]

This report first compares certain regions with their renewable energy output and evaluates the trends in the data over a given time frame. It will then progress on to looking to see if GDP impacts this trend or whether other factors contribute to renewable energy share such as politics or geographical location.

II. METHODOLOGY

Data sets were first analysed from the World Bank Group and the International Energy Agency (IEA).[2][3]

The data set from the IEA includes the energy produced by renewables in TOE (tonne of oil equivalent). This measures the energy produced and directly compares this with the amount of oil that would be used to produce the same amount of energy. This creates a direct comparison with oil, making the data more meaningful and directly infers the benefit of renewable energy, as it shows how much oil renewable energy has replaced.

The rest of the project primarily looks at the percentage of renewable energy produced across a region/country. This report chooses to consider hydroelectric power as a renewable energy source. The data set provided excludes hydroelectric power as a renewable. Therefore, I have added the hydroelectric power to the already existing renewable energy percentage to calculate a total renewable percentage.

There is also debate as to whether nuclear energy should be considered renewable. This report considers nuclear energy as its own separate category. When nuclear energy is being referred to within the report, this will be made clear.

The data set provided by the World Bank Group separates each country into 4 income groups. These are high income, upper middle income, lower middle income and

low income. The GDP required for each of these categories is set by the World Bank Group. The GDP of countries is measured in United States Dollars (USD).

The graphs produced within the results section below are produced with python packages such as Pandas, NumPy and Matplotlib.[4][5][6]

III. RESULTS

A. Renewable Energy Trends In Specific Regions

Firstly, trends in renewable energy production were reviewed. In Fig. 1 below, three regions (East Asia & Pacific, Europe & Central Asia and North America) have their renewable energy production plotted between 2000 and 2018.

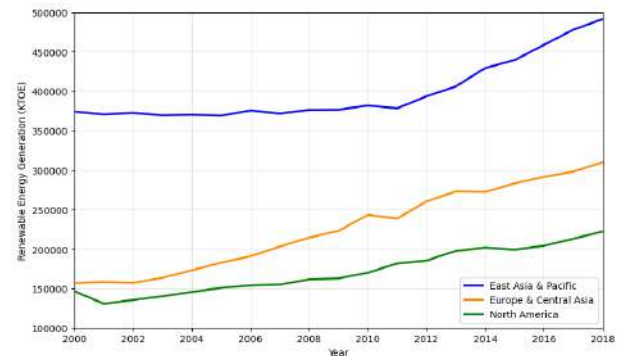


FIG. 1. Renewable Energy Production (2000-2018)

Fig. 1 shows that East Asia & Pacific's renewable energy generation is significantly greater than the other two regions. This could be due to geographic location, but on examining this more closely, this does not appear to be the case. The difference appears to be driven by politics.

For example, the Europe & Central Asia plot shows a steady increase which likely reflects EU policies for climate change such as the European Climate Law. The European Climate Law has deadlines that the EU must legally hit in a certain time frame. One of these laws is that the EU must be completely net zero by 2050. [7] These deadlines are likely to contribute to the gradual increase in renewable energy generation as the EU pushes towards this goal.

This is somewhat similar to the policies in North America as well as East Asia & Pacific. One of the major contributors to the increase that begins in 2011, as shown in Fig. 1, is the change in Chinese policies. As China is state owned, they are able to make decisions and act on them quickly and efficiently.

An example of this was their decision to increase energy security, which they have done in recent years by investing massively into renewable energy, especially wind power. This is due to state-led investment into renewable energy which has caused a massive increase in wind turbine manufacturing.

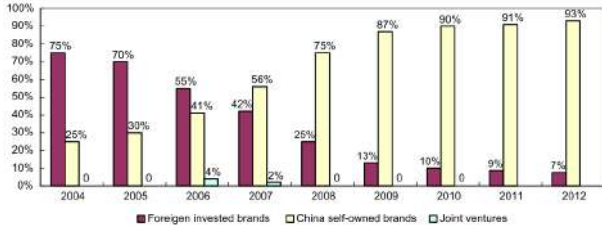


FIG. 2. Market shares of international and China's domestic wind turbine manufacturers.[8]

As Fig. 2 suggests, there has been a massive increase in the manufacturing capacity of wind turbines within China from a 25% market share in 2004 to a 93% share in 2015. This is a direct result of the state-led initiative and this has allowed China to scale its renewable energy production quickly. China currently accounts for 72% of the global market for wind turbines as of 2024.[9]

This large increase in manufacturing output has allowed China to increase their renewable share quickly, as manufacturing is less expensive than in other regions of the world.

This data is further reinforced if you look at the renewable percentage of the regions as in Fig. 3 below.

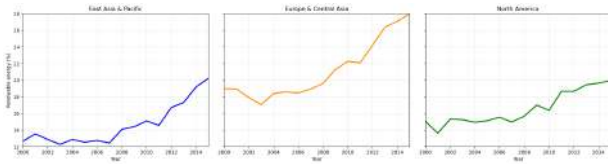


FIG. 3. Regional Trends in Renewable Energy (% of total), 2000-2015

Fig. 1 suggests that East Asia & Pacific has significantly more renewable energy generation than the other regions. However, Fig. 3 shows that the percentage of their total renewable energy is lower than the other regions. Europe & Central Asia have a much higher percentage of renewable energy. East Asia & Pacific and Europe & Central Asia have both shown an increase in renewable energy whereas North America has not prioritised renewable energy.

B. Political Impacts on Energy Production Types

Another consideration is whether a country's income determines what energy source they invest in. A key example of this is nuclear energy which is often considered to be more expensive than other sources of energy. The World Nuclear Association (WNA) suggests that the real cost of nuclear power is in the construction of the power plant rather than the fuel used to power the station; potentially suggesting that access to nuclear energy is only possible for higher to upper middle income countries.[10]

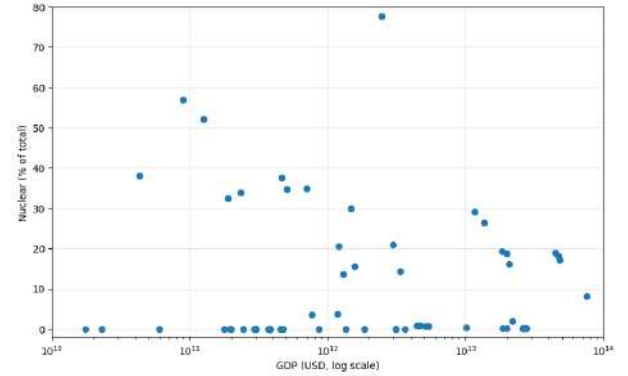


FIG. 4. GDP against Nuclear (% of total) for 2015

Fig. 4 compares the GDP of all countries in the data set with its nuclear percent in 2015. Given what the WNA suggested about the cost of these facilities, it was expected that a positive correlation would be seen with an increase in GDP. However, as Fig. 4 shows, this is not the case. Instead, it shows that there is no correlation between GDP and nuclear percentage.

Instead, the nuclear percentage is determined by government beliefs and political stances. For example, France has the highest nuclear production by percentage of any country at around 78% according to the data set.[3] This can be seen in Fig. 4 as the highest point on the graph.

The French government strongly believes that nuclear energy is the best option for them and as a result has invested heavily into this energy sector.[11] Thus proving that the primary energy source of a country is not purely determined by GDP or geographical location, but also by political perspectives. This applies to all other sources of energy, where opinions and beliefs determine which electrical production method is selected.

C. Impacts of Income on Energy Production Types

Finally, correlation between a country's income and its energy production methods was considered. It was hypothesised that a higher income country would have more renewable energy or nuclear power as they would have the necessary funds to invest in these areas. This was briefly looked at in Fig. 4 where it was discovered that actually this was not the case.

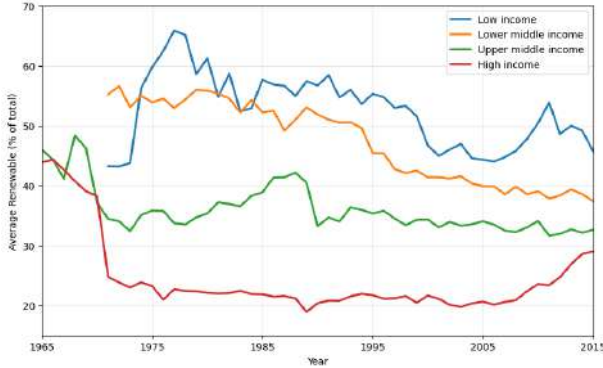


FIG. 5. Renewable Energy Share by Income Group Over Time (1965-2015)

Fig. 5 also supports this point. For this graph, an average was calculated for all countries in each income group for each year. This graph confirms that on average the high income countries have the lowest percentage of renewable energy, with low income countries having the highest percentage.

The reason these countries have a lower percentage of renewables is likely the same reason they are high income. Most of the countries that have exported or continue to export fossil fuels end up being the high income countries, as these are a sought after commodity. An example of this is the United States, which is the largest oil producer in the world and as of 2024 has 22% of the global share of world oil production.[12] But this only contributed to 8% of the country's GDP in 2024.[13]

It may not necessarily be the case that the export of fossil fuels currently makes countries have a high GDP. High income countries tended to export fossil fuels earlier than other countries. The United States economy was built on oil just as the United Kingdom economy was built on coal. These countries are now high income as they had the money to invest in new areas to generate more income. They built non-renewable power stations as these were less expensive as the resources required to fuel the stations were more freely available and at the time there was less environmental awareness.

This is also true for the United Arab Emirates, whose economy was also built on oil but in more recent years.

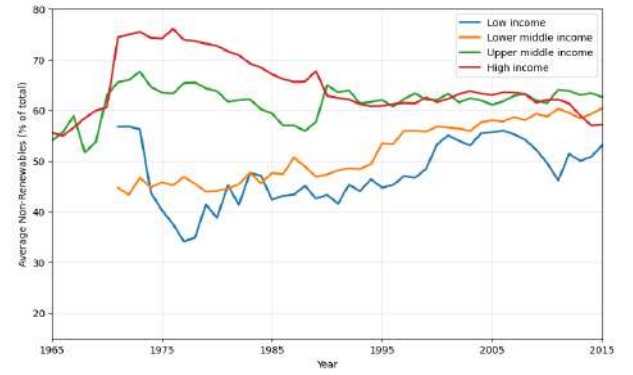


FIG. 6. Non-Renewable Energy Share by Income Group Over Time (1965-2015)

Fig. 6 further confirms this hypothesis and again shows the correlation between income and renewable energy usage. This time the graph only looks at oil, coal and natural gas as non-renewables excluding nuclear energy, but the correlation is still clear.

The plot of high income does show renewable energy production start to increase in recent years as non-renewables decreases. This could be due to ensuring energy security or societal pressures.

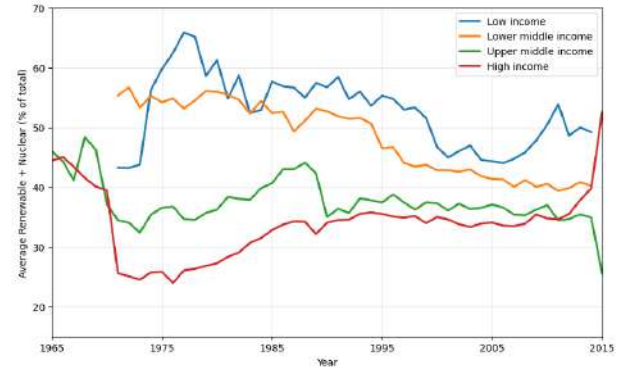


FIG. 7. Renewable and Nuclear Energy Share by Income Group Over Time (1965-2015)

Fig. 7 includes renewable and nuclear energy and shows a significant spike in the last few years for higher income countries suggesting that high income countries are investing heavily into nuclear energy.

This does suggest that nuclear energy is more accessible to higher income countries, but as mentioned earlier, although this is true, it is often more dependent on government ideology.

The spike in renewable and nuclear energy in high income countries, in Fig. 7, is most likely due to a decrease in the cost of renewable energy technologies over time and a decrease in coal production in high income countries.

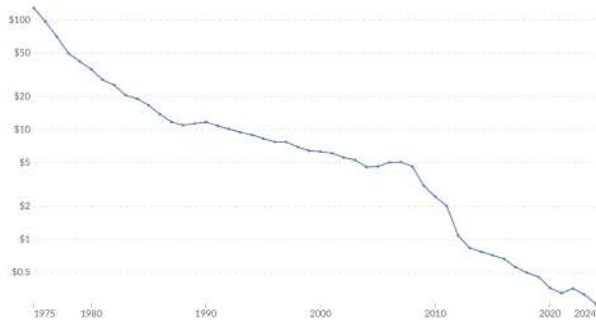


FIG. 8. Solar photovoltaic panel prices (1975 - 2024)[14]

For example, there has been a decrease in solar photovoltaic panel prices, as shown in Fig. 8. This is one of the reasons why renewable energy production has begun to increase as renewable technologies have become more affordable and can be scaled up more easily. Likewise, as renewable technology continues to improve as research continues, wind turbines will become more efficient, making them more viable for large scale wind farms.



FIG. 9. Coal Production (1900 - 2024)[15]

Another contributing factor is the collapse of coal production in high income countries as seen in Fig. 9. Less coal production in high income countries makes using coal power more expensive as coal has to be imported, thus encouraging high income countries to move to renewable or nuclear power production. This does not hold true for oil extraction as there is significantly more profit in oil than other fossil fuels.

Both are examples of why renewable and nuclear en-

ergy has increased as it is more affordable and scalable than ever before. However, each country and region have different policies and opinions on energy production, which is the main contributor to which energy source is used.

IV. CONCLUSION

Overall, GDP has a strong correlation with the type of energy production used, but the relation is not as straightforward as it first seems. There are many different aspects to consider from geographical location, income and politics both locally and globally.

Initially, it was hypothesised that the geographical area played an enormous role in the energy source utilised. This was proven to be partially true, with high income countries becoming high income due to their early exports of fossil fuels such as coal and oil. However, these countries no longer export fossil fuels to the same level as they have invested in other income streams. This, as a result, provides them with the means to invest in whatever energy production method they see fit.

So, although GDP does play a role in which energy source is used, this is not necessarily the primary reason. When analysing certain energy production methods such as nuclear, it was found that there is no real correlation between GDP and nuclear percentage. This is due to the policies and opinions of the individual countries. This is especially true for nuclear power as it is a more controversial source of energy since various nuclear incidents, such as Chernobyl and Fukushima. Political beliefs were found to be one of the greatest determiners of which energy sector is selected.

This is also the case when you consider global energy trends such as renewable energy technologies becoming less expensive, a greater push from the general public for renewable energy, and a collapse in coal production in high income countries.

Suggestions to improve this report would be to expand the data set to earlier years and ensure all the data in the data set is correct and complete. There are currently a lot of missing values in the data set that have to be removed when calculating the averages for the plots. This skews the data, especially for low income countries, as these are the countries with most missing values and therefore, are not represented as accurately as others.

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